# WORKCRAFT Local Master Blueprint

**PROPRIETARY GO ARCHITECTURE & ZERO-COST ENGINE SPECIFICATION**

## 1. Executive Summary & Core Architecture

WORKCRAFT Local delivers an enterprise-grade, managed WhatsApp automation engine tailored for salons and parlors in Pakistan and the MENA region. By replacing legacy Python and third-party SaaS tools with a custom compiled **Go (Golang)** microservice utilizing the native `whatsmeow` WebSocket protocol, the platform completely eliminates third-party per-message fees and software subscription costs.

| $0.00 | < 10ms | 98%+ |
|---|---|---|
| Per-Message Meta Charge | Go Engine Latency | Net Operating Margin |

## 2. Platform Comparison Matrix

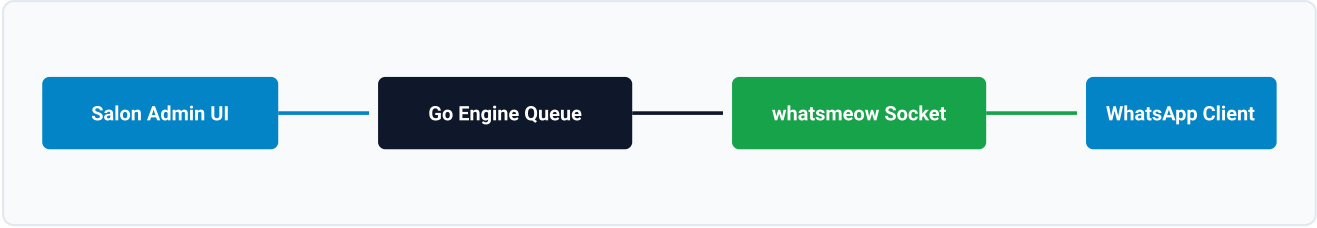
| System Vector | Legacy Python / SaaS | Official Meta Cloud API | WORKCRAFT Go Engine |
|---|---|---|---|
| **Messaging Cost** | $15–$99/month sub | PKR 13.20 per marketing blast | $0.00 Free Socket |
| **Execution Speed** | Interpreted (Slow) | Network dependent | Compiled Native Go |
| **Security / Leakage** | High risk (Interpreted code) | Third-party trust | AES-256 + RLS Isolated |
| **Onboarding** | Complex API keys | Meta Business Verification | 1-Scan QR Pairing |
| **Ban Risk** | High if unthrottled | Zero (Official) | Safe Rate-Limited Queue |

## 3. Security & Data Protection Framework

The system utilizes **Supabase (PostgreSQL) Row-Level Security (RLS)** to guarantee complete tenant isolation. Client phone lists, booking records, and video broadcast logs for Salon A are physically impossible to access from Salon B. The compiled Go binary hides source code entirely, offering zero attack surface for external code injection.

## 4. System Data Flow & Ban-Safety Shield

To prevent WhatsApp account suspensions during Day-1 200+ contact video broadcasts, the Go backend runs a randomized throttling queue. Messages are dispatched individually with artificial delays (4s to 9s), simulating authentic human interaction.




## 5. Implementation Roadmap & AI Prompt Specifications

To construct this exact application visually inside AI tools such as v0.app or Bolt.new, copy and execute the master prompt provided below.

| Phase | Milestone | Core Deliverable | Status |
|---|---|---|---|
| **Phase 1** | Go Core Engine | WebSocket QR Pairing + Supabase Schema | Ready for Dev |
| **Phase 2** | Ban-Safety Shield | 4s–9s Throttled Video Queue + Opt-Out Engine | Ready for Dev |
| **Phase 3** | Mobile Front-End | 375px Mobile Portal + Review Booster | Ready for Dev |
| **Phase 4** | Revenue Control | Automated 2-Day Billing & Red Alert Lockout | Ready for Dev |

## 6. Master Prompt for AI Studio / v0.app / Bolt.new

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Build a mobile-first (375px) SaaS portal for WORKCRAFT Local using React, Tailwind CSS, Go backend structure, and Supabase. Features required: 1. QR Code Pairing Tab: Generate a live WebSocket connection QR code for salon onboarding. 2. Contacts Manager: CSV upload tool for 200+ contacts with opt-out status flags. 3. Video Broadcast Console: File uploader for 60-second MP4 deal videos. Include a visual rate-limiting queue progress bar showing messages sending every 4 to 9 seconds. 4. Tri-Lingual Flow Builder: State machine visualizer for English, Arabic, and Urdu routing. 5. Revenue Control: Maintenance payment tracker with a 'Red Alert' account suspended banner for overdue fees. Style: Clean executive UI, deep slate (#0f172a) and ocean blue (#0284c7) accents, no crop issues on mobile.
```